

Report



SPDC-BFPC UPSTREAM GAS SUPPLY, GATHERING, EVACUATION & PROCESSING (UGS-GEP) PROJECT

Process Simulation Study - Gas Processing Plant

Prepared for:

Brass Fertilizer & Petrochemical Company Limited

Doc. No. EA/SPDC-BFPC UGS-GEP/PRC-RPT-004-A01

Endorsed by:

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28th February 2017

Statement of Confidentiality

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SUMMARY PAGE

Client: BRASS FERTILIZER & PETROCHEMICAL COMPANY LIMITED	
Project: SPDC-BFPC UGS-GEP	Title: Process Simulation Study – Gas Processing Plant
Facilities: Gas Processing Plant	Doc No.: EA/SPDC-BFPC UGS-GEP/PRC-RPT-004

The administration of this document is in respect of the Hysys Simulation of the Gas Processing Plant aspect of the SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project.

The document defines the basic design concepts, issues, and considerations in the subject matter of process simulation, and the result process design and product yields that will form the basis of the project's Front End and Engineering Design (FEED).

It is also an input & accompaniment to the conceptual study and basic design reports & presentations to DPR as procedural and statutory requirements in the process of securing regulators' approval to progress to the FEED / Detail Design that will lead to the project's FID and implementation.

The report has relied and used data and reference documents provided by client, and if further information becomes available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEY WORDS: BASIS FOR DESIGN, PIPELINE, PIPELINE MANIFOLD PLATFORM, GAS PROCESSING PLANT, ROW

REFERENCES	
Document No.	Description
TO-DAA-00001	Basis for Design
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Gas Gathering & Surface Facilities Concept Screening Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-001	Process Simulation Study – Hysys - Pipelines
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Process Simulation Study – Pipesim - Pipelines

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Abbreviations

The Following definitions and abbreviations have been used in this document.

Organizational

AKAB	Akabel
BBOU	Bubouwe
BFPC	Brass Fertilizer Petrochemical Company Limited
EA	Epic Atlantic Limited
ELEP	Elepa
NUNR	Nun River
OPUG	Opugbene
SEIB	Seibou
SPDC	The Shell Petroleum Development Company of Nigeria Limited

Technical Terms

B/L	Bulkline
CF-MH	Central Field MF Header
CSU	Condensate Stabilization Unit
CTP	Custody Transfer Point
CW	Cooler Water
E-MF	Evacuation Manifold
EOS	Equation of State
F-MF	Field Manifold
GEP	Gathering, Evacuation and Processing

GSPA	Gas Supply and Purchase Agreement
NGL	Natural Gas Liquids

Technical Terms (Cont'd)

NAG	Non Associated Gas
OML	Oil Mining License
OSD	On-Stream Date
PR	Peng-Robinson
SMF	Satellite Manifold
T/L	Trunkline
UGS	Upstream Gas Supply

Units of Measurements

km	Kilometre
m	Metre
M	Thousand
MM	Million
MMscfd	Million Standard Cubic Feet per Day
MTPD	Metric Tonnes Per Day
°C	Degree Celsius
bar	Bar

1 Administration

This Process Simulation Study is in respect of the SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project. It stems from the Concept Select Study on the Gas Gathering Concept, and presents the selected design concept as basis for the Basic Design of the project.

It is also an input and accompaniment to the Conceptual Study & Basic Design Memoranda to be presented to DPR as statutory requirements in the process of securing regulators' approval to execute the FEED / Detail Design that will lead to the project's FID and implementation.

1.1 Validity & Change Management

The report has relied and used data and reference documents provided by client. Some of the data are preliminary, based on limited information, but to a large extent, however, are adequate for this phase of work: - conceptual study and basic design.

For example, analogue PVT data was used in the absence of actual PVT, and preliminary pipeline distances were estimated based on desktop route selection study. These design data, and the consequent simulation results will invariably need to be updated during FEED, by which time actual PVT data would have been obtained, and the FP survey done to establish actual distances and elevations of the pipeline routes.

Similarly, the project timelines as set out in the execution schedule was predicated on FID date of June 2018, and that will correspondingly change with actual FID date.

Also, the security situation in the region with its wide spate of vandalism may dictate that certain design provisions be changed during the FEED. The situation will continue to be monitored, and if still perceived to be a cog in the wheel of innovation, some of the ideal designs will be replaced with what will be practically suitable for the situation.

For example, in as much as it is feasible to lay cable from the IPP to the remote manifold platforms to transmit electricity to motorise the platform valves in the Manifold Platforms Power Supply Options, or install solar systems on the platforms, there will be no need to implement such options if the facilities (cables & solar systems) will eventually be vandalised in no time. Instead, it will be rather more appealing to limit the process option to manual valve operations, without the features of valve motorisation that will require electricity.

Epic Atlantic therefore reserves the right to update accordingly, the opinion set out in this report, as further information and / or refinement become available.

1.2 Background & Introduction

The SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project is constituted by Brass Fertilizer & Petrochemical Company Limited (BFPC), in partnership with The Shell Petroleum Development Company of Nigeria (SPDC).

The project scope includes Gas Gathering Pipelines, and a Gas Process Processing Plant that will provide processed gas as feedstock to a Petrochemical Plant and a Power Plant. The Gas Processing Plant will process natural gas gathered from five SPDC fields – Bubouwe Bou, Elepa, Nun River, Seibou, and Opugbene - in OMLs 33, 32, & 36.

The Gas Plant will comprise three trains, (each with a processing capacity of 150 MMscfd) and condensate storage capacity of 50,000 Tonnes. In this respect, BFPC have concluded the Gas Supply & Processing Agreement (GSPA) with SPDC for 270 MMscfd of the natural gas feedstock. The Plants (Gas Plant, Petrochemical Plant, and Power Plant) will all be located in the same complex at Akabel, Brass Island, Bayelsa state of Nigeria, near the shores of Atlantic Ocean.

The Petrochemical Plant will comprise a Methanol Plant of 1.75 MTPA capacity, and a Urea Plant of 2.6 MTPA total capacity, built in two phases of 1.3 MTPA each. It will also have storage capacity for 150,000 Metric Tonnes of Methanol and 100,000 Metric Tonnes of Urea. The facilities complex will also comprise a captive export jetty of 4 berths for up to 35,000 DWT vessels.

The current on-stream date for the project is December 2020, and to achieve that, FID is planned for June, 2018. In pursuit of these business objectives, BFPC have virtually concluded Project Management & EPC contracts with EIL, Saipem, and TCC, who are internationally renowned contractors in the implementation of the subject projects type.

The Gas Processing Plant, and the associated Gas Gathering System / Pipelines which will transport wellhead gas from the SPDC fields to the Gas Processing Plant, constitute the key upstream enabler facilities for the realisation of the objective supply of processed gas supply as feedstock to the Petrochemical Plant for the product manufacturing of methanol & urea, and to the Power Plant for power generation.

1.3 Scope of Document

The scope of the document covers the Process Simulation Study with Hysys and its evolution of results for phase-1 gas production processing.

✚ Phase - 1 Gas Production: BBOU and ELEP Field

Process simulation for the subsequent phases will be addressed during FEED.

2 Basis and Assumptions

Base Assumption

The Gas Plant Processing Facility temperature ambient medium was taken as 25 °C.

Software

The Gas Plant process simulation study was undertaken using Aspen Hysys v7.3.

Water saturation extension was explored to determine the saturated water in the gas.

Simulation Property Package

The thermodynamic equation of state (EOS) used for the process simulation is; Peng-Robinson (PR) fluid property package.

Gas Sources and Schedules for Plant Operation Over 25 Years (2020-2045)

The feed gas sources and schedules for the Gas Processing Plant operations; is as highlighted in the production forecast below.

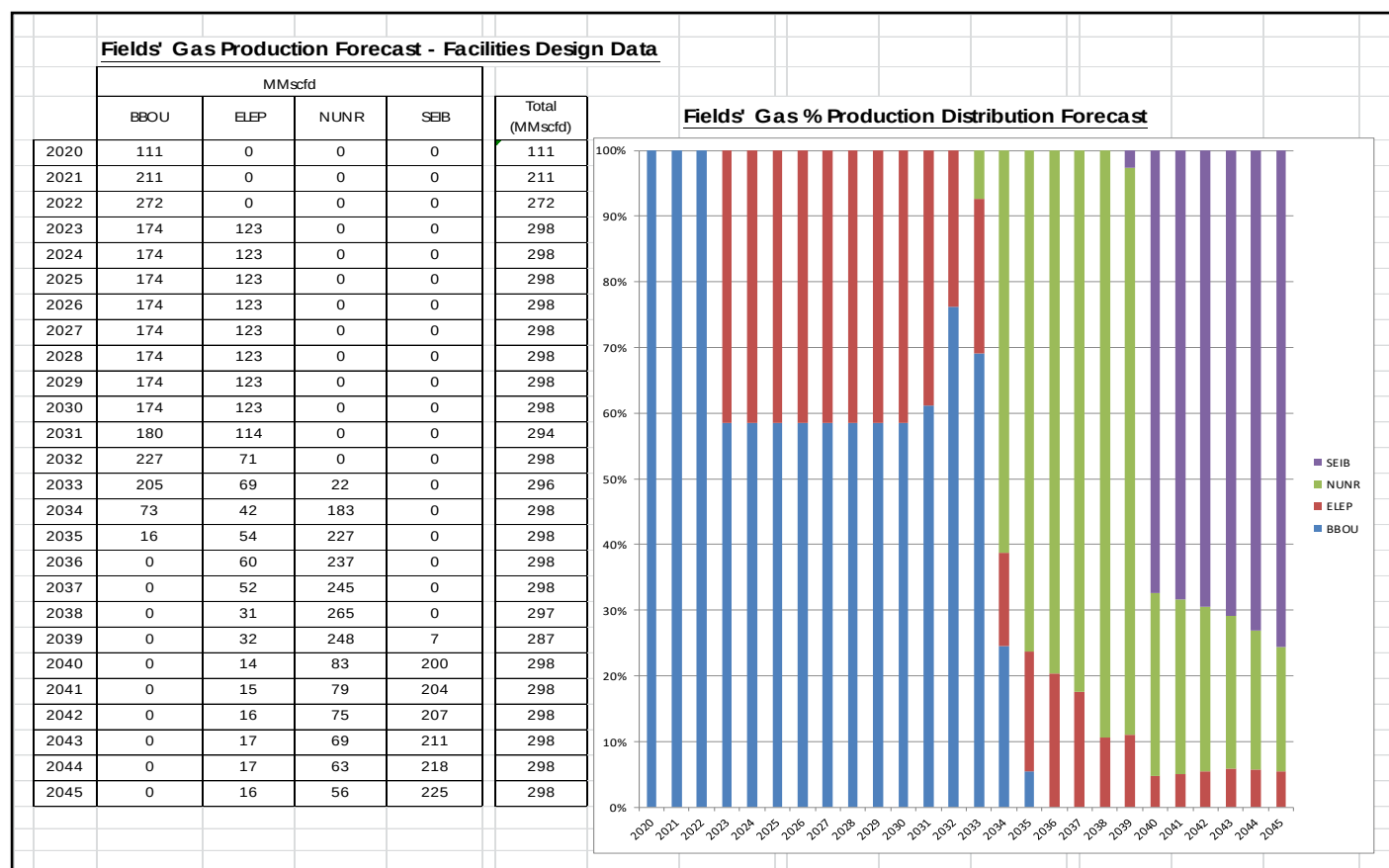


Figure 2.1: Production Forecast

Field's Data

Table 2.1 shows BBOU, ELEP, NUNR, SEIB, and OPUG field reservoir conditions and fluid composition in mole fraction basis as provided by SPDC.

Similarly, table 2.2 shows water saturated compositions of BBOU, ELEP, NUNR, SEIB, and OPUG field compositions.

Table 2.1: BBOU, ELEP, NUNR, SEIB, and OPUG Reservoir Compositions & Conditions

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Table 2.2: BBOU, ELEP, NUNR, SEIB, and OPUG Water-Saturated Reservoir Compositions

Saturated Gas Compositions																					
	BBO U			ELEP			NUNR			SEIB			OPUG								
Component	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid						
N2			0.07			0.00			0.00			0.07			0.10						
CO2			0.59			2.17			0.77			1.53			2.21						
H2S			0.00			0.00			0.00			0.00			0.00						
C1			92.61			79.87			72.37			77.84			67.24						
C2			3.15			6.00			7.63			5.87			7.88						
C3			1.01			2.53			4.82			3.37			4.76						
iC4			0.61			0.73			1.45			1.21			1.94						
nC4			0.44			0.79			1.96			1.37			1.06						
iC5			0.25			0.44			0.94			0.82			1.15						
nC5			0.19			0.32			0.76			0.57			0.68						
C6			0.23			0.64			1.02			0.91			1.06						
C7+			0.53			4.31			7.65			5.48			10.86						
H2O			0.34			2.19			0.63			0.95			1.08						
Total			100.00			100.00			100.00			100.00			100.00						

3 Project Scope and Battery Limits

3.1 Field Gathering Facilities up to AKAB Gas Plant

- a. Starts from: Field manifold (MF) of Shell at each field
- b. Field gathering facilities include:-
 - Manifold (S-MF) at NUNR (18" x 10 m) and manifold (E-MF) at ELEP (12" x 10 m)
 - Bulk flow lines connecting field MF with NUNR S-MF or ELEP E-MF per schematic diagram provided (20" x 20 km, 16" x 9 km)
 - Bulk flow line connecting NUNR S-MF with ELEP E-MF (24" x 45 km)
 - Bulk flow line connecting ELEP E-MF with the gas plant at Akabel (24" x 28.5 km)

3.2 Gas Plant – BFPC Scope

1. Gas plant starts from: Outlet of 24" bulk flow line entering
2. Gas plant scope includes:-
 - a. Common for Both trains: Inlet Surge Drum Separator
 - b. Per train:
 - Feed gas separation Unit,
 - Condensate stabiliser unit,
 - Gas drying unit (TEG or MEG – to be confirmed during engineering phase),
 - Propane chiller and Refrigeration unit (provisional - to be confirmed later during engineering phase),
 - Turbo-expander and NGL recovery unit,
 - Fractionation unit containing demethanizer and debutanizer,
 - NGL storage bullets and pumps (common for all trains),
 - Condensate storage and pumps (common for all trains)
 - Utilities and off-sites (common for all trains)
 - Gas plant battery limit at:-
 - Gas custody transfer at inlet flange of gas pipelines going to BFPC and power plant (flange within gas plant battery limit, B/L - the gas supply pipelines not included).
 - Condensate custody transfer at outlet of the transfer pumps supplying to pipeline and/or truck loading facility (assumed, as no details provided).
 - NGL custody transfer at outlet of NGL transfer pumps supplying to NGL pipeline feeding jetty loading facilities – the pipeline not included as jetty details not known.

4 Description of the Proposed Process Configuration

In developing the process configuration for the gas plant, only two parallel trains are considered to be built at this stage and provision is kept for adding a third train of similar capacity in future. Refer to attached Schematic Flow Diagram which is briefly described below

4.1 Inlet Gas Separation Unit

Raw natural gas from all remote manifolds (MF) along with all the associated condensate and water is routed to the gas processing plant at Akabel via ELEP E-MF.

The incoming gas-liquid stream will be received in Surge Drum-Separator to separate out bulk of the free condensate and water present in the stream. Depending on the water volume, 3-phase separation arrangement may be applied in the drum – separated water will be routed to waste water treatment/disposal system.

Condensate will be sent to the Condensate Stabilizer unit. The gas at about 65 barg pressure and 50 - 55 °C temperature will be split in two separate streams feeding the two trains separately. In each train the gas is cooled to about 35 - 40 °C either in an aerial cooler or in a shell & tube cooler using recirculating cooling water (CW) and the outlet gas is passed through HP Gas K.O. Drum to knock out the condensed liquid.

The outlet gas from HP Gas K.O. Drum will be passed through a fine coalescing type Filter-Separator to separate out fine droplets and mists of liquid and solid particles from gas, which may otherwise deactivate the downstream TEG Drying unit.

4.2 Condensate Stabilization

The raw condensate coming from Inlet Gas Separation area will be received in Stabilizer Feed Surge Drum which will separate out most of the remaining free water and will feed the stabilizer column at a pre-set steady rate.

The stabilizer column will have one bottom a reboiler and two side-reboilers. The stabilizer column (stripper) strips off the dissolved light gases (C1/C2, H₂S, inert gases etc.) to meet the desired specification, usually RVP= 10 - 11, water content < 500 - 1000 ppm and H₂S and C₄ - limits).

The hot condensate leaving the upper side-reboiler is cooled to about 35 - 50 °C in an aerial (or a CW cooler) and then stored in two condensate storage tanks in the tank farm pumping to pipeline to Jetty for loading.

4.3 Gas Drying or Dehydration

In this case the optimal drying process would be to use TEG drying which is best suited for bulk drying of product gas, about 7 lb water/MMscf dry gas (-50 to -60 °C dew point at atmospheric pressure) for NGL recovery by auto-refrigeration or chilling to around -32 to -35 °C (dew point below -32 °C at operating pressure downstream of the turbo-expander). Alternately, mole-sieve dryers which can dry to a much lower dew point (< -90 °C) could be used, but may prove to be costly, considering the requirement for shallow NGL recovery of -30 to -32 °C. TEG contactor will be either bubble-cap tray tower or CS or a packed tower using SS structured or random packings.

The wet TEG exiting the bottom of the contactor column will pass through a plate-frame type exchanger to exchange heat with hot TEG from the regenerator drum and will be fed at the top of the regenerator/stripper. The regenerator reboiler will be heated by recirculating hot oil.

4.4 Gas Chilling and Refrigeration

Provision for a packaged propane chilling units (1+1 per train) is kept in design, but may not be installed in this phase of the project (subject to confirmation during the design stage). Each package will consist of a compressor, filter and a cooler (aerial or water cooled). Propane will be recirculating in the process unit under its own pressure where it will chill the dry gas by evaporating. The LP vapor will return back to the compressor.

The hydrocarbon gas will be chilled in 2 stages in a plate & frame, or a shell & tube, type exchanger. The chilled gas will enter into a cold liquid K.O drum where the condensed liquid will be separated (mostly C4 and C5+).

4.5 Turbo-expander for NGL Recovery

The outlet gas from the drum at below -1 to 1 °C will enter turbo-expander unit where the expander drops its pressure to about 34 - 35 barg pressure.

The gas chills down to below -30 to -32 °C by auto-refrigeration. A JT valve is installed parallel to the expander. The outlet gas from turbo-expander goes to Expander Cold Liquid Separator where the condensed liquid, mostly C3/C4 and some C2, will separate out.

The separated gas is recompressed by re-compressor which is coupled in tandem with the Expander on the same shaft.

The power developed by the expander drives the compressor. The outlet stream containing liquid and gas is routed to Expander Cold Separator where bulk of the lean gas (mostly methane with some ethane) comes out through the top of the vessel and is passed through a series of cross-heat exchangers (mentioned before) to recover the cold and itself gets warmed up.

The warmed gas at about 30 - 40 °C and 34 - 38 barg pressure will go to product pipelines for distribution to BFPC and Power Plant.

4.6 NGL Fractionation (Demethanizer / Debutanizer)

The liquid (raw NGL/C5+) from the bottom of the Cold Liquid Separator and other drums are routed to Demethanizer column where most of the remaining methane is stripped out from the liquid streams.

The exiting gas from Demethanizer, after exchanging their cold with various warm streams (refer to flow diagram) is mixed with bulk of the gas in the product gas pipeline. The reboiler of the column is heated with hot oil.

The liquid from bottom of the Demethanizer (Demethanized - NGL) is routed to Debutanizer column to separate out C5+ components from product NGL (C3/C4). The product NGL comes out from top of the column through the over-head cooler and the reflux drum. C5+ Plus stream comes from the bottom of Debutanizer column as product. The reboiler is heated by hot oil.

4.7 Condensate Storage and Dispatch

The stabilized condensate is store in a blanketed cone roof tank. Provision for an additional tank is made in the layout to accommodate future extra condensate. The condensate is pumped out to ship loading jetty.

4.8 NGL Storage and Dispatch

The product NGL is stored in two bullets (alternately sphere may be provided, but may not be optimum as the expected volume of product NGL will be low to moderate). The NGL will be pumped into a pipeline going to a ship loading jetty.

4.9 Utilities and Offsite Facilities

- The necessary utility systems like hot oil, instrument air, fuel gas system, recirculating cooling water system (if CW is used for bulk cooling) will be provided in the utility block from where they will be distributed to the process units.

- A HP flare will be provided in a suitable remote location of the plant which will be designed for emergency flaring.

4.10 Wellhead Manifolds / Bulk Flowlines

The manifolds and the flowlines are proposed to be of CS with extra corrosion allowance (6 - 10 mm) mainly to deal with wetCO₂ corrosion. In our opinion, no SS or Duplex SS material will be required and they may not be any superior as there may be other associated problems, besides exorbitantly high cost of SS/duplex materials.

Drying at MF sites is also not practical, as it will involve separation of high pressure liquid hydrocarbons from the well fluid at MF sites and then the associated problems of handling them (either separate pipelines of SS or SS tankers plus pumps – both do not appear good solutions). The most practiced way of handling well fluid containing CO₂ at high pressure is to use CS pipes with adequate corrosion allowance.

Also, in this case, each gas source/reservoir (like BBOU, ELEP etc.) and corresponding MFs are projected to operate only for 10 - 15 years. Therefore, design with corrosion allowance will be optimal.

In addition, the flow lines and the MFs should have corrosion inhibitor injection facility, if adequate inhibitor is not already added to well downhole to protect well tubes/casing etc.

Corrosion coupons/electrodes should be installed on the lines to monitor corrosion rates, mainly near the inner-bottom side of the flowlines where water mostly collects, and to adjust inhibitor injection dosage. The lines should be periodically pigged with scrapper pigs to sweep out stagnant water if collected at low points in the pipeline. However, this should be firmed up during the detail design taking into account the preference and experience of the owner / operators. The flare K.O. drum and pump will be provided in a suitable location close to the flare to knock out any liquid and return back to the plant.

5 Observations and Improvement Suggestions

We have the following observations and initial comments:

5.1 3 - Trains VS 2 - Trains

It seems; it is optimal and most cost effective to have only two trains, each of 150 MMscfd capacities, instead of three trains of same capacity. Since maximum processing capacity required at any time during the lifetime of the plant (25 years) will be within 300 MMscfd, the third train is difficult to justify at this stage, unless it is an owner's preference.

In gas plants of this nature there is no need to have an exclusive standby train - only redundancy in critical equipment should be enough to ensure high plant reliability and availability.

5.2 Gas Drying at Manifolds Sites

This is not essential and may not be an optimal solution to circumvent CO₂ corrosion problem. CO₂ corrosion occurs in presence of liquid water saturated with CO₂, which can occur even at insignificantly low CO₂ concentration in gas.

This high pressure CO₂ corrosion problem arises almost in all oil / gas production facilities everywhere. Considering that presence of liquid water and CO₂ in pipelines is unavoidable, the most optimal solution adopted in most places is use of CS with extra corrosion allowance (6-10 mm) and periodic corrosion inhibitor injection into the pipelines at well side manifolds.

The bulk flowlines may also need periodic pigging (about once in 3 - 6 months) to sweep out any pocket of collected stagnant water at low points in the pipelines. Use of SS 304 / 316 or duplex steel may be superfluous (and prohibitively expensive) in most cases.

5.3 Need for Gas Sweetening & CO₂ Reduction

There is no H₂S present in the feed gas (per gas composition) and also the CO₂ concentration is low enough to affect the product gas specification required by the consumers, mainly the power plant (LHV usually above 900 - 920 Btu/scft).

Fertilizer plants usually have no problem with CO₂. Hence, need for gas sweetening unit is questionable (subject to further study), unless the owner opts to invest on this for extra security. However, modular sweetening units may be added later, if found necessary later based on actual composition of feed gas obtained from the reservoirs.

5.4 NGL Recovery Unit; Deep VS Shallow Recovery

As feed gas streams contain low amount of C₂/C₃, deep recovery (chilling below -30 °C) to capture high amount of C₂ (> 50% of total C₂) may not be economically justified. In this case, we may recommend chilling down to only about -32 °C, which will keep most of the critical equipment within the limit of normal LTCS steel (above -45 °C). This will avoid going to SS material for major equipment in NGL unit.

6 Further Considerations and Reviews

6.1 Preliminary Plot Plan

A preliminary typical plot plan is developed based on the limited data available at this stage and making various assumptions. This may significantly change should the actual site conditions, basic data, product specs and the process configuration change later due to optimization/detail design. The present layout is good only for information and planning at this stage.

6.2 Utility Services and Chemicals Requirement

It is assumed that all required utility services like raw water, electricity (grid) etc., are locally available. The raw water will be treated at site to meet CW make-up, plant water and domestic water requirement, fire water system make-up etc.

The main/bulk chemical requirement is for MDEA and TEG. This should be procured in advance and sufficient stock will be kept all the time at site to meet the normal requirement – necessary provisions of tanks are made in the layout. Besides, some small amount of water treatment chemicals, antifreeze chemical like methanol, antifoaming agents etc. shall be kept at site which will be supplied mostly in drums.

6.3 Preliminary List of Effluents and Emission

There is no major effluent or emission is expected from the plant. The main liquid effluent (continuous) is small amount of produced water from the Inlet filter-separators, condensate stabilizer and sweetening units.

Gas emission is mainly the acid gas from the regenerators of the Sweetening units, which will be mainly CO₂ (no H₂S is envisaged in the feed gas). Also, very small amount of vent gas from the dryer stripper-regenerator which may be vented to atmosphere. Both these streams are expected to have no impact on environmental regulations.

6.4 Indicative Cost of the Gas Processing Plant as Proposed

A ballpark estimate of installed cost based on experience on similar plants in North America should be around US \$ 250 - 300 million (today's dollars) with about +50% / -40% level of accuracy. This estimate is roughly for equipment and facilities as listed in Annexure-1 below. This is assuming at least 50 - 60% of engineering and equipment are supplied from East Asia.

In absence of firm labor, construction and transportation costs and applicable duties, taxes, environmental / regulatory requirements in Nigeria, the estimate is likely to change significantly later. A more precise Class-III (+25% / -20%) type estimation may be possible after completion of FEED stage of engineering.

6.5 CAPEX Profile for the Gas Processing Facilities,

Bearing in mind:-

- a. 2 x 150 MMscfd processing facilities trains are required for the start
- b. Indication by the production forecast that the supply gas is very wet later in project life than in the first fourteen (14) years.

Projected CAPEX profile would be approx. around 75 - 80% during initial project engineering and construction stage (2 - 3 years before commissioning). Major expenditure will be in 2 gas processing and NGL recovery trains and in common (shared) facilities, like fractionation (DC1 / DC4), utilities, off-sites, storage, loading facilities, onsite developments etc. all of which will be required at the beginning.

The condensate volume expected even at the initial stage (BBOU and ELEP) would require two condensate trains from the beginning (ELEP liquid is shown as high as NUNR and SEIB). The same trains will be able to handle the condensate volume expected at later stages. Only a few equipment can be deferred and added in stages (subject to detail engineering findings), like 1 (out of 3) inlet drum and separator, 1 (out of 3) refrig unit, may be 1 NGL bullet, some transfer pumps and a few other minor equipment.

6.6 Turndown Rates for the Gas Processing Facilities

Typical turn down of each train will be about 40% of their design capacity (mainly dictated by tray / packed towers, like TEG contactors, regenerators, DC1, DC4, condensate stabilizers). Since there are two 50% trains, each of 150 MMSCFD, the overall turndown on plant capacity of 300 MMSCFD will be $40\% \times 50\% \times 300 \text{ MMSCFD} = 60 \text{ MMSCFD}$. If the production rate for a sustained period is between 60 and 150 MMSCFD, only one train may be operated to optimize operating costs.

6.7 Condensed + Produced Water Treatment Package and Associated Cost

Since the inlet gas / well-fluid analysis does not show any H₂S, the condensed and produced water is not expected to be sour. Hence no major facility like sour water stripper will be required for treatment before disposal.

Mainly an API separator (gravity type), set of cartridge or media filters, disposal pumps and small chemicals dosing skids (antiscalant, biocide / anti-biofouling agent etc. depending on disposal type) may be required. The type and specification of the pump will depend on where and how the waste water will be disposed of - to surface streams / rivers etc. or to underground injection.

The entire unit may not be a major cost item and may be covered under the overall indicative project cost.

6.8 Liquid Holdup Profile in the Gathering and Evacuation System

Liquid hold-up will depend on equilibrium between gas and liquid phases which is a function of line size, gas and liquid densities and superficial velocities. This can be predicted only during detail design by pipeline simulation (if included in engineer's scope), line sizing, ground contour, temperature change etc.

In this case, the flow will be mostly uniform mixed phase and in design the line velocity will be chosen mostly in the stratified flow regime to avoid slug formation / problem in the lines. The liquid hold up will be minimum in this case and the collected liquid will be evacuated by periodic pigging, based on actual field experience.

6.9 Shared CSU Unit between Trains VS Dedicated Unit for each Train

The condensate volume will be quite significant from the beginning all through the plant life-cycle. To handle this large volume, single CSU (condensate stabilization unit) train may be too large from practical points, like construction (diameter will be large), cost of single stabilizer vessel, shipping & transportation, erection, operation, efficiency etc. Also individual train with interconnection may be the best choice considering wide turndown ratio required to handle changing condensate volume with time.

(For information, at the initial stage of correspondence with Toredos/PSI in June '16 the condensate figure was given quite low and only for BBOU. We had then suggested a common / shared condensate stabilization unit – please refer to initial block flow diagram. But at the present high condensate figures from all sources, it may be optimal to consider an individual stabilizer per train. However, the same may be studied further in detail design stage)

List of Major Units and Equipment in Gas Plant

(For order of magnitude cost estimation)

Unit Name [Major equipment in each train]	No. of Trains
Inlet Gas Separation & Filtration Unit [Inlet Surge Drum-Separator, Feed Gas Cooler, HP Gas K.O. Drum, Filter-Separator]	1 (or 2) (Common for both process trains) (can also be separate for each train, per owner's preference)
TEG Dryer Package Unit [TEG contactor, drums, regenerator-stripper, vessels, heat-exchanger, pump, piping, valves, instruments etc.]	2
Chilling & NGL Recovery Unit [Coolers, refrig units (3 -common for both trains), chillers, heat-exchangers, drums, Expander-re-compressor, anti-fridge injection package etc.]	2 (Refrig units 3 nos. may be common for both trains)
NGL Fractionation Unit [DC1, DC4 Columns, exchangers, reboilers, drums, vessels, pumps etc.]	2
Condensate Stabilizer & Tank Farm [Stabilizer columns, reboilers, drums, exchangers, pumps, NGL bullets (2 or 3 common), Condensate tanks (2 common) etc.]	2
Hot Oil System Package [2 or 3 units common for both trains, complete with drums, pumps	2 or 3 (Common for both process trains)
Utility and Offsite Units [Instrument and Plant air, fuel gas, cooling water (if CW is selected for cooling), inert gas, plant water treatment, waste water treatment and disposal by injection or land streams (details & type to be decided later), condensate and NGL loading facilities]	1 Common
Fire Water System [Tanks, pumps, headers, monitors, sprinklers, hydrants etc. – FW may be required per recommended code as large volume of highly flammable condensate and NGL will be produced, handled and stored on site – to be confirmed during design]	1 Common
Piping, valves, racks, instruments, controls for all non-package units, plant buildings, civil works within plant etc.	
DCS, control room, instrument-shelters, electrical sub-stations, electrical distribution etc.	
Not Considered in Present Scope / Estimate: Power generation, except an emergency power set Field pipelines and facilities Supply/receiving pipelines outside plant battery limit Gas distribution compressor, if required Initial site and infrastructure development etc.	

6.10 List of Equipment

List of major equipment and units are shown on the attached Plot Plan and Equipment Layout.

Attachments:

Schematic Flow Diagrams (5 sheets)

-  Inlet Separation Unit
-  Gas Drying Unit
-  Chilling and NGL Recovery Unit
-  Fractionation Unit
-  Condensate Stabilisation and Tank Farm
-  Gas Plant Water Treatment & Disposal

7 Appendix

7.1 Appendix A: Gas Processing Plant PFS; PFD and Plot Plan

S/N	Deliverable Description	File
A1	PFS AKAB Gas Plant	 PFS AKAB Gas Plant
A2	PFD AKAB Gas Plant - Inlet Sep System	 PFD AKAB Gas Plant - Inlet Sep System
A3	PFD AKAB Gas Plant - Gas Drying System	 PFD AKAB Gas Plant - Gas Drying System
A4	PFD AKAB Gas Plant - Chilling & NGL Recovery System	 PFD AKB G/P Chilling & NGL Recovry Sys.
A5	PFD AKAB Gas Plant - NGL Fractionating System	 PFD AKB G/P - NGL Fractionating Sys
A6	PFD AKAB Gas Plant - Condensate Stabilisation System	 PFD AKB G/ P - Cond Stabilisation Sys
A7	PFS AKAB Gas Plant Waste Water Treatment & Disposal	 PFS AKAB G/P Water Treatment & Disposal
A8	AKAB Gas Plant - Plot Plan & Equip Layout	 Gas Plant _Plot Plan & Equip Layout

7.2 Appendix B: PFS, Hysys PFD and Datasheets

S/N	Deliverable Description	File
B1	Hysys PFD	 Hysys PFD AKAB Gas Plant
B2	Gas Plant Feed Datasheet	 Gas Plant Feed
B3	Train-1 Gas Feed Datasheet	 Train-1 Gas Feed
B4	Train-1 _ Product Lean Gas Datasheet	 Train-1_Product Lean Gas
B5	Train-1 _ De-Ethanizer _ C1/C2 Gas Datasheet	 Train-1_DC2_Gas
B6	Train-1 _ De-Ethanizer _ C3+ Condensate Datasheet	 Train-1_DC2_C3+ Condensate
B7	Train-1 _ De-Butanizer _ C1/C2 Gas Datasheet	 Train-1_DC4_Gas
B8	Train-1 _ De-Butanizer _ C3/C4 Distillate Datasheet	 Train-1_DC4_C3/C4 Distillate
B9	Train-1 _ De-Butanizer _ C5+ Condensate Datasheet	 Train-1_DC4_C5+ Condensate
B10	Train-1 _ Cooled C5+ Condensate _ Tank Feed Datasheet	 Train-1_Cooled_C5+ Cond Tank Feed
B11	Train-1 _ SD + Stabilizer Gas Datasheet	 Train-1_SD + Stabilizer Gas
B12	Train-1 _ Stabilizer C3+ Condensate Datasheet	 Train-1_Stabilizer C3+ Condensate
B13	Train-2 Gas Feed Datasheet	 Train-2 Gas Feed
B14	Train-2 _ Product Lean Gas Datasheet	 Train-2_Product Lean Gas

B15	Train-2 _ De-Ethanizer _ C1/C2 Gas Datasheet	 Train-2_DC2_Gas
B16	Train-2 _ De-Ethanizer _ C3+ Condensate Datasheet	 Train-2_DC2_C3+ Condensate
B17	Train-2 _ De-Butanizer _ C1/C2 Gas Datasheet	 Train-2_DC4_Gas
B18	Train-2 _ De-Butanizer _ C3/C4 Distillate Datasheet	 Train-2_DC4_C3/C4 Distillate
B19	Train-2 _ De-Butanizer _ C5+ Condensate Datasheet	 Train-2_DC4_C5+ Codensate
B20	Train-2 _ Cooled C5+ Condensate _ Tank Feed Datasheet	 Train-2_Cooled_C5+ Cond tank Feed
B21	Train-2 _ SD + Stabilizer Gas Datasheet	 Train-2_SD + Stabilizer Gas
B22	Train-2 _ Stabilizer C3+ Condensate Datasheet	 Train-2_Stabilizer C3+ Condensate

8 Key References

Reference-1: Gas Gathering & Evacuation Pipelines HYSYS Simulation Report

	Document Ref	Document	Information
Ref 1	EA/SPDC-BFPC UGS-GEP/PRC-RPT-001 Gas Gathering & Evacuation Pipelines HYSYS Simulation Report	 EA SPDC-BFPC UGS GEP PRC RPT 001	Gas Gathering & Evacuation Pipelines HYSYS Simulation Deliverables

9 List of Tables

Table 1: BBOU, ELEP, NUNR, SEIB, and OPUG Reservoir Compositions & Conditions

Table 2: BBOU, ELEP, NUNR, SEIB, and OPUG Water-Saturated Reservoir Compositions

Table 3: Raw gas sources and schedules for plant operation over 25 years (2020 - 2045)

Table 4: List of Major Units and Equipment in Gas Plant

10 List of Figure

Figure 1: Production Forecast